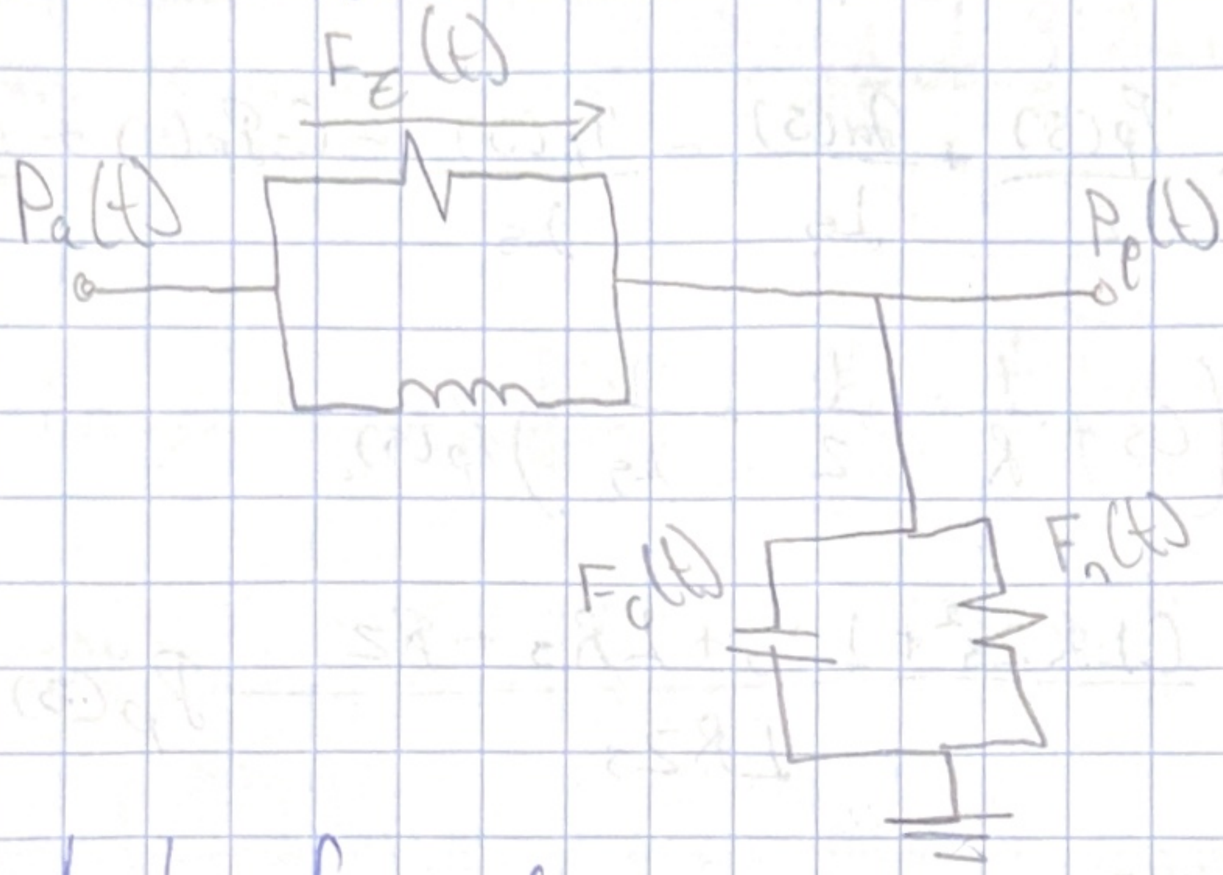
Modelado de sistemas Fisiologicos 07/10/2020

Analisis matematico



Función de transferencia

Ecuaciones principales

$$F_a(t) = F_2(t) + F_2(t) = F_c(t) + F_R(t)$$

$$F_2(t) = \frac{P_a(t) - P_p(t)}{2}$$

$$F_L(t) = \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt$$

$$F_c(t) = C\frac{dP_p(t)}{dt}$$

$$F_R(t) = \frac{P_p(t)}{R}$$

Transformada de laplace

$$F_a(s) = F_2(s) + F_L(s) = F_c(s) + F_R(s)$$

$$F_2(s) = \frac{P_a(s) - P_p(s)}{2}$$

$$F_L(s) = \frac{P_a(s) - P_p(s)}{Ls}$$

$$F_c(s) = Cs\,P_p(s)$$

$$F_R(s)\ \frac{P_p(s)}{R}$$

$$\frac{P_a(s) - P_p(s)}{2} + \frac{P_a(s) - P_p(s)}{Ls} = Cs P_p(s) + \frac{P_p(s)}{R}$$

(2)

Procedimiento algebraico

- $$\frac{V_a(s)}{2} - \frac{V_p(s)}{2} + \frac{V_a(s)}{Ls} - \frac{V_p(s)}{Ls} = CsV_p(s) + \frac{V_p(s)}{R}$$

- $$\left(\frac{1}{2} + \frac{1}{Ls}\right)V_a(s) = \left(Cs + \frac{1}{R} + \frac{1}{2} + \frac{1}{Ls}\right)V_p(s)$$

- $$\frac{Ls + 2}{2Ls}V_a(s) = \frac{CLR2s^2 + L2s + LRs + R2}{LR2s}V_p(s)$$

$$\frac{V_p(s)}{V_a(s)} = \frac{\frac{Ls + 2}{2Ls}}{\frac{CLR2s^2 + L(2+R)s + R2}{LR2s}}$$

Resultado

$$\frac{V_p(s)}{V_a(s)} = \frac{LRs + R2}{CLR2s^2 + L(2+R)s + R2}$$

(3)

Error en estado estacionario

Este error se calcula cuando la entrada $\mathcal{L}[P_a(s)]$ es un escalon unitario, es decir, $P_a(s) = 1/s$:

$$e(s) = \lim_{s\to 0} s P_a(s)\left(1 - \frac{P_b(s)}{P_a(s)}\right) = \lim_{s\to 0} \frac{1}{s}\left(1 - \frac{LR_s + R_2}{CLR_2 s^2 + L(2+R)s + R_2}\right) = 0$$

$$e(t) = 0v$$

Estabilidad del sistema en Lazo abierto

Para analizar la estabilidad en Lazo abierto se deben calcular dos polos:

$$CLR_2 s^2 + L(2+R)s + R_2 = 0$$

$$a = CLR_2$$
$$b = L(2+R)$$
$$c = R_2$$

$$\lambda_{1,2} = \frac{-L(2+R) \pm \sqrt{L^2(2+R)^2 - 4CLR^2Z^2}}{2CLR_2}$$

Hipotenso

$Z = 0.010$
$L = 0.001$
$R = 1.000$
$C = 3.500$

$$\frac{P_a(s)}{P_a(s)} = \frac{LR_s + R_2}{CLR_2 s^2 + L(2+R)s + R_2} = \frac{0.001s + 0.01}{3.5\times10^{-3}s^2 + 0.00101s + 0.01}$$

$$\lambda_1 = -14.429 + 8.8051i,$$

$$\lambda_2 = -14.429 - 8.8051i$$

(4)

Por lo tanto, el sistema presenta una respuesta estable y sobreamortiguada

Normotenso

$Z = 0.010$
$L = 0.004$
$R = 1000$
$C = 1.500$

$$\frac{P_p(s)}{P_a(s)} = \frac{LR_s + R_2}{CLRZs^2 + L(Z+R)s + R_2} = \frac{0.001s + 0.01}{1.5\times10^{-5}s^2 + 6.0061s + 0.01}$$

$\lambda_1 = -12.062$

$\lambda_2 = -55.272$

Por lo tanto presenta una respuesta estable y sobreamortiguada.

(5)

Hipertenso

$Z = 0.010$
$L = 0.001$
$R = 1.000$
$C = 1.000$

$$\frac{V_o(s)}{V_i(s)} = \frac{LR_2 s + R_2}{CLR_2 s^2 + L(2+R)s + R_2} = \frac{0.001s + 0.01}{0.00001 s^2 + 0.00101 s + 0.01}$$

$\lambda_1 = -10.0$
$\lambda_2 = -10.0$

Por lo tanto, el sistema presenta una respuesta estable y críticamente amortiguada.

(6)

Modelo de ecuaciones integro-diferenciales

$$F_2(t) + F_L(t) = F_C(t) + F_R(t)$$

$$\frac{P_a(t) - P_p(t)}{2} + \frac{c}{L}\int\left[P_a(t) - P_p(t)\right]dt = C\frac{dP_p(t)}{dt} + \frac{P_p(t)}{R}$$

Nota: La unica variable desconocida es la salida del sistema.
$P_a(t)$ NO se puede despejar de términos en los que se esté derivando o integrando

$$\frac{P_a(t)}{2} + \frac{1}{L}\int\left[P_a(t) - P_p(t)\right]dt - C\frac{dP_p(t)}{dt} = \frac{P_p(t)}{R} + \frac{P_p(t)}{2}$$

$$P_p(t)\left(\frac{1}{R} + \frac{1}{2}\right) = \left[\frac{P_a(t)}{2} + \frac{1}{L}\int\left[P_a(t) - P_p(t)\right]dt - C\frac{dP_p(t)}{dt}\right]$$

$$P_p(t)\frac{Z+R}{RZ} = \left[\frac{P_a(t)}{2} + \frac{1}{L}\int\left[P_a(t) - P_p(t)\right]dt - C\frac{dP_p(t)}{dt}\right]$$

$$P_p(t) = \left[\frac{P_a(t)}{2} + \frac{1}{L}\int\left[P_a(t) - P_p(t)\right]dt - C\frac{dP_p(t)}{dt}\right]\frac{RZ}{Z+R}$$